



(51) International Patent Classification:
G05B 23/02 (2006.01)

(21) International Application Number:
PCT/US2017/045685

(22) International Filing Date:
07 August 2017 (07.08.2017)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:
62/380,653 29 August 2016 (29.08.2016) US

(71) Applicant: SIEMENS CORPORATION [US/US]; 170
Wood Avenue South, Iselin, New Jersey 08830 (US).

(72) Inventors: DASGUPTA, Arindam; 23 Brownstone Drive,
Avon, Connecticut 06001 (US). KULKARNI, Anand
A.; 9420 Ardrey Woods Drive, Charlotte, North Carolina
28277 (US). CHAKRABORTY, Amit; 2 Inverness Lane,
East Windsor, New Jersey 08520 (US).

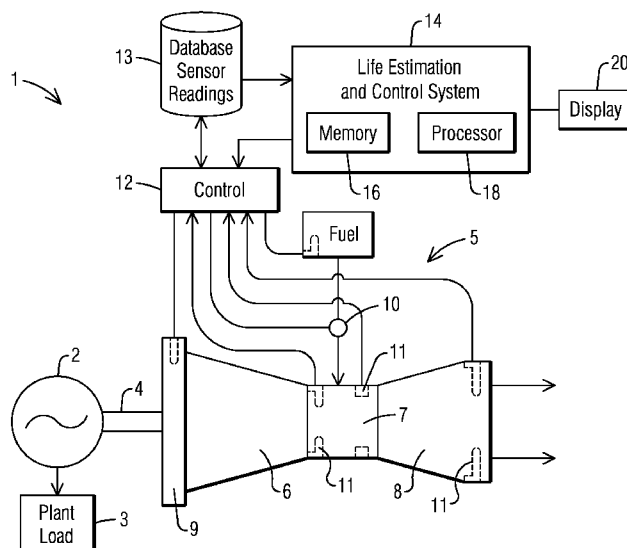
(74) Agent: FIL, Michele S.; 3501 Quadrangle Blvd. Suite. 230,
Orlando, Florida 32817 (US).

(81) Designated States (unless otherwise indicated, for every
kind of national protection available): AE, AG, AL, AM,
AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ,
CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO,
DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN,
HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP,
KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME,
MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ,
OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA,
SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN,
TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

(84) Designated States (unless otherwise indicated, for every
kind of regional protection available): ARIPO (BW, GH,
GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ,
UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ,
TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK,
EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV,
MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM,

(54) Title: MACHINE LEARNING-AIDED MODEL-BASED METHOD FOR ESTIMATING GAS TURBINE COMPONENT LIFE

FIG. 3



(57) Abstract: A method and system for estimating a life consumption for a component of a gas turbine engine. The method includes acquiring and storing data relative to the current operating condition of the component during operation of the gas turbine engine. The acquired data is then validated. Based on the acquired data, the life consumption of the gas turbine component is estimated. A method of assessing a gas turbine engine is also provided which is accomplished using the life estimation of the components measured in the method for estimating the life consumption for a component.

TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW,
KM, ML, MR, NE, SN, TD, TG).

Published:

— *with international search report (Art. 21(3))*

MACHINE LEARNING-AIDED MODEL-BASED METHOD FOR ESTIMATING GAS TURBINE COMPONENT LIFE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application Serial No. 62/380,653, filed August 29, 2016, and herein incorporated by reference.

BACKGROUND

1. Field

[0002] This disclosure relates generally to component life prediction, assessment, and evaluation of components. In particular, a method is presented for estimating gas turbine component life.

2. Description of the Related Art

[0003] Estimation of component life in operating gas turbines is a complex and extremely challenging problem which has very significant impact on the operations, availability, reliability, economics and safety of power plants. Since the gas turbine components are exposed to harsh conditions and a wide range of temperatures, pressures and corrosive environments, different components have different life expectancy at different sites. Since every time a component needs to be replaced, the machine becomes unavailable for generation, there is a fine line between running the component close to its safe and useful life and yet avoiding a forced outage due to component failure.

[0004] There are several methods currently used for estimating a component's life. In general, these methods can be classified into four broad approaches: a) Model-Based approach, b) Service-Based approach, c) Statistical/Probabilistic-Based approach, and d) Soft Computing approach.

[0005] The model-based approach involves empirical, analytical or numerical

model creation in order to calculate the component's life. Depending on how the model is created, external sources are needed to accompany the estimation such as the component's geometry, material data, and engine operating and health conditions. Non-destructive testing (NDT) and destructive testing (DT) should be carried out using conventional techniques when the life fraction is greater than 0.5 (50%). Creep life estimation using the total life approach can be done by using either a life-based model, a strain-based model, or a damage-based model.

[0006] The service-based approach involves damage evaluation and remaining life assessment of the service exposed component, requiring direct access to the components, which means that the machine needs to be taken out of service. The present status of the component's material is positioned within the standard scatter band either by measuring its properties, hence providing a refined prediction, or by direct assessment of the extent of the damage experienced by the component as a result of the actual service exposed. The main methods of assessing the remaining component life will involve both non-destructive tests (NDTs) and destructive tests (DTs).

[0007] Although a statistical approach has been partially used in some of the approaches, most of the approaches use statistics in order to assist the model development which in general is deterministic in nature. In a statistical/probabilistic approach, statistical/probabilistic theory is used as the main lifing method. This approach is used either as a means to substitute the existing model, or as a means to account for uncertainties in the influencing parameters or variables that will affect the creep life behaviour, or to provide the basis of performing failure risk analysis.

[0008] Although the model-based and service-based approaches have been successfully used over the years, limitations and complexity of the approaches have driven researchers to use another form of life estimation approach. This is particularly important for modern machines as they operate closer to material limits and multiple mechanisms of damage interact towards failure, making simple models focusing on a single mechanism inadequate. For this reason, soft computing techniques such as ANN (artificial neural network), fuzzy logic and evolutionary algorithm have been used to capture the complex, ill-understood interactions. Among these soft computing

approaches, ANN has been predominantly used.

[0009] The approach presented here recognizes that given the complexity of the problem, all the methods described above need to be effectively combined for accurate predictions that can be commercially deployed.

SUMMARY

[0010] Briefly described, aspects of the present disclosure relate to a method and system for estimating the life consumption of a turbine component. Additionally, the disclosure relates to a method for assessing the current state of an operating gas turbine.

[0011] A method for estimating life consumption for a component of a gas turbine engine is provided. Data is acquired and stored relative to the current operating condition of a gas turbine component during operation of the gas turbine engine. The acquired data is then filtered and validated. At this point in the method, the life consumption of the component may be estimated.

[0012] A method for assessing the current state of an operating gas turbine is also provided. The method as described above is performed for each of a plurality of gas turbine components that are cooperatively integrated on an operating gas turbine engine. The current state of the engine is assessed using a statistical combination of the life consumption of each of the plurality of gas turbine components.

[0013] A system for estimating the life consumption of a plurality of gas turbine components in an operating gas turbine engine is also provided. The system includes a processing unit, the processing unit including a program storage device and a processor. The processor is in communication with a database that is used to store acquired data and is configured to be queried in real time to access the stored acquired data. The program storage device embodies in a fixed tangible medium a set of program instructions executable by the processor to perform the steps of: acquiring and storing data relative to the current operating condition of the components during operating of the gas turbine engine, validating the acquired data, and determining life

consumption of each component based on the acquired data. The system also includes a control module having a processor communicatively coupled to the processing unit and configured to change a physical parameter of the gas turbine engine based on the life consumption of the components. A display is configured to display the determined life consumption of each component.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] Fig. 1 is a flowchart of the proposed method of estimating a life consumption for a gas turbine component,

[0015] Fig. 2 is a flowchart of the proposed method of assessing the current state of a gas turbine engine, and

[0016] Fig. 3 is a diagram of an exemplary gas turbine indicating inclusion of sensors used to measure the performance of the turbine, and illustrating a life estimation and control system.

DETAILED DESCRIPTION

[0017] To facilitate an understanding of embodiments, principles, and features of the present disclosure, they are explained hereinafter with reference to implementation in illustrative embodiments. Embodiments of the present disclosure, however, are not limited to use in the described systems or methods.

[0018] The components and materials described hereinafter as making up the various embodiments are intended to be illustrative and not restrictive. Many suitable components and materials that would perform the same or a similar function as the materials described herein are intended to be embraced within the scope of embodiments of the present disclosure.

[0019] In an embodiment, a first method is presented in which a life consumption of a gas turbine component in an operating gas turbine is estimated. A flowchart of an

embodiment of the first method may be seen in Fig. 1. In a further embodiment, a second method in which the current state of an operating gas turbine engine is assessed. A flowchart depicting the second method may be seen in Fig. 2.

[0020] In a first step of the first method, data is acquired and stored 100 relative to the current operating condition of the gas turbine component(s) of the gas turbine engine. The data may be collected from a deployed gas turbine. The 'component condition' represents the data and information available from conformal on-line continuous monitoring sensors and standard engine supervisory sensors for detailed operational history.

[0021] There are two types of data that can be acquired, Discrete Event Data and Continuous Operating Data (COD). Event data include all historical data and records of the targeted gas turbine engines, such as history of faults, breakdowns, repairs, overhauls, start/stops etc., while COD may include all measured thermodynamics observations along the gas path and mechanical data, indicative of its current operating and health condition. However, event data are not usually acquired in an automated manner and therefore, manual data entry may be required. COD are normally collected by using available instrumentation on gas turbine engines, for example, vibration measurements through accelerometers, velocity pick-ups and displacement sensors; pressure measurements at various gas path stations of the gas turbine engines through pressure transducers; temperature measurements through thermocouples and/or resistance temperature detectors; shaft speed measurement; corrosion measurement through a corrosion probe; and ambient conditions measurements, for example, ambient pressure, temperature and relative humidity. It may be understood that other measurements may be taken by various sensors used to monitor the performance of a gas turbine.

[0022] Sensor measurements obtained from a gas turbine may contain random errors. In addition, biases may be present in the measurements due to incorrect calibration and/or sensor faults. The random noises and biases in the measured data give rise to inaccuracy in estimates of health parameters. To avoid this inaccuracy, the measured data should be validated before they are used.

[0023] At this step, the acquired measurements are filtered to reduce measurement noise and validated 200, through appropriate sensor validation techniques. A good summary of state of the art noise filtering techniques and related signal processing processes is given in Vachtsevanos G., '*Intelligent Fault Diagnosis and Prognosis for Engineering Systems*', John Wiley, New York, 2006. In an embodiment, the acquired data is filtered in order to remove random errors by averaging multiple data points collected over a certain time period. The filtered data may then be validated. Some validation techniques aim at detecting and isolating any possible sensor fault, while other validation techniques also allow an estimation of the underlying sensor bias. In an embodiment of the proposed method, Artificial Neural Networks (ANN) such as described in Romessis C., Mathioudakis K., 2003, '*Setting Up of a Probabilistic Neural Network for Sensor Fault Detection Including Operation with Component Faults*', ASME Journal of Engineering for Gas Turbines and Power, Vol 125, No. 3, July 2003, pp. 634-641, will be used to detect sensor anomalies and biases.

[0024] Once this data is filtered through extensive data quality checks and safeguards, the data is processed to develop required boundary conditions for components and failure mechanisms, execute algorithms to establish component operational history lifing parameters (e.g., temperature and stress histories) which are then fed into damage mechanism lifing algorithms for estimation of consumed (and remaining) life. The algorithms and entire process are calibrated and verified with non-intrusive component sensor data, observed component condition, inspection and outage data, repair data, and fleet data. The process flow involves data gathering and management, data processing, assessment and predictions followed by a feedback loop.

[0025] The ultimate goal of gathering data using advanced sensors and applying sophisticated analytics is to assess the current state of the engine (or fleet of engines) and predict the future condition and requirements based on historical and statistical use.

[0026] Engine components are affected by several different internal and external factors as they operate including local environmental conditions, fuel quality, and electrical demand patterns in the market served by the machine. During this process

the engine and its components sustain different types and degree of damages induced by different well-known mechanisms. In that sense, these damages can be construed as ‘life consumption’ of the engine. In typical ‘hot section’ components these damage mechanisms may include corrosion, erosion, oxidation, hot corrosion, creep, low cycle fatigue, thermomechanical fatigue and high cycle fatigue. A complete and detailed assessment of the life consumed (or damage done) for each interval the machine operates and each start/stop cycle over long periods of operation is impossible.

[0027] Assessment 300, 400 can be simplified by adopting a machine-learning model, the ‘box-model’, where the damage done to engines by different operating modes may be subdivided into fatigue mechanisms and duration of operation mechanisms. Then, it is appropriate to subdivide engine operating modes into equivalent starts which accounts for all the fatigue damage, and equivalent baseload hours for the duration based life usage. With this subdivided counting system nearly all circumstances of gas turbine operation can be covered. The process results in two sets of equations for each component as follows:

Equivalent Start Counters for cyclic damage including LCF (low cycle fatigue) and TMF (thermo-mechanical fatigue)

$$N_{ES}^{cg} = \sum_{i=1}^{N_{cycles}} F_{transient}^{(ES,cg)} \quad (1)$$

Where:

N_{ES}^{cg} is the total number of cycles of all types (including start-stop and loading sub-cycles),

cg refers to specific component or component group, and

$F_{transient}^{(ES,cg)}$ refers to the transient event start factor.

[0028] Damage over each complete cycle is estimated as a % of life consumed based on models described in literature and internal experience and is integrated over the in-service time of a particular component.

Equivalent Life Consumed

$$\begin{aligned}
 & LC_m^{total} \\
 & = F_m^{online} \sum_{i=1}^{N_{online}} LC_{online} \Delta t_{online} + F_m^{offline} \sum_{i=1}^{N_{offline}} LC_{offline} \Delta t_{offline} \\
 & + F_m^{start} \sum_{i=1}^{N_{start}} LC_{start} \Delta t
 \end{aligned} \tag{2}$$

Where:

LC_m^{total} is the total life consumed for damage mechanism ‘m’,
 LC_{online} , $LC_{offline}$, LC_{start} are life consumed during online, offline and start conditions determined by laboratory tests and physics-based modeling as a % of useful life,
 Δt are intervals of operation and
 F_m^{online} , $F_m^{offline}$ and F_m^{start} are correction factors developed using machine learning techniques to operational data as applied to specific field conditions of individual machines.

[0029] Thus, the life consumption of each component may be computed based on a two term approach. Then, from a statistical combination of duty/load cycles of the engines and power/temperature conditions of all the components assessed, a life estimation of the gas turbine engine may be performed 400. A flowchart illustrating the second method of assessing the current state of the gas turbine may be seen in Fig. 2. From this assessment, a reliability of the gas turbine engine may be estimated.

[0030] The different damage mechanisms are assessed individually during operation and based on individual-use cases, maintenance decisions are formulated and implemented. The damage mechanisms considered include corrosion, erosion, 2oxidation, hot corrosion, creep, HCF (high cycle fatigue), and off-line spinning. The challenge and the success of the process depend on the estimation of the ‘life consumption factors’ and the ‘correction factors’ for the engine. Considering the large number of variables and range of these variables that are used as inputs into the physical models for estimation of these factors, computational complexity, and

expense of the experimentation that will be required to comprehensively account for all conditions, a surrogate model approach for factor estimation is proposed.

[0031] In the proposed method, a surrogate model may be used instead of analytical models as they may be too detailed and take too much time to update and execute in real-time when considering all the pertinent variables and their combinations that may affect the gas turbine engine. Moreover, analytical models may not be completely accurate when used for real conditions. Hence, the creation of a surrogate model that relies on a physical model but may be adjusted through use of coefficients that are derived from operational data is proposed. A surrogate model may be quickly updated without revisiting detailed analysis every time conditions in the system change such that the surrogate model may be quickly updated through re-evaluation of the correction factors should they change as the conditions change or, for example, as the gas turbine ages.

[0032] A surrogate model may be used to estimate the correction factors utilized in the equations (1) and (2) as described above, for example, the correction factors F_m^{online} , $F_m^{offline}$ and F_m^{start} used in equation (2). In the context of the proposed component life estimation method, correction factors are those quantities that correct the theoretical formulations to real-life operation quantities and are unique to each machine. The large dimensionality of the problem and input variation may be addressed using dimensional reduction techniques such as Auto Encoders (AE) and Artificial Neural Networks (ANN), or other Machine Learning methods that reduce engineering computation significantly. In certain classes of problems there are dimensions that have a low-impact on the objective function. Identifying low-impact dimensions is very important because it can reduce the size/complexity of the problem. Auto-Encoders are modern Machine Learning algorithms that can identify hidden structure in data, such as correlated variables and are powerful in dimensionality reduction. ANNs can provide a nonlinear, physical model-free, multivariate surface data fitting learning algorithm that can automatically determine and learn the underlying functional relationship between inputs and outputs directly from data without a hypothesized functional form. Additionally, ANNs are suitable for adaptation to parallel computation architecture and have good generalization

capabilities (universal approximators). They are also suitable for incremental learning, enabling the neural network models to be improved incrementally as new data become available or change previously calculated weight functions as input data changes (for example, if an engine starts to be operated differently as demand curve changes).

[0033] A surrogate model (which is really a set of correlations or a regression model) can be interrogated repeatedly and cheaply with engine operational data (time series data) to calculate cumulative life consumption – a real-time or almost real-time assessment method. The process will be also used predictively by using past patterns of engine use and projecting into the future. Thus, the surrogate model may be utilized to predict the life consumption of each gas turbine component. From predicted life consumption of each gas turbine component, the life consumption of the gas turbine engine itself may be predicted 500.

[0034] The overall predictive tool development needs to be validated at two levels. The first level is a validation of the results predicted by the deterministic tools that correlates to the sensor data (observed data) in the laboratory and the material degradation analysis. This is required because the surrogate model needs a training dataset that is larger than what a laboratory testing plan can economically provide. So, a large part of the training dataset will be provided from physics-based computational models which must be verified and adjusted themselves using a sampling method. The second level of validation needs to accomplish a correlation between data generated in a controlled laboratory environment and observations from the field with their associated uncertainties. Surrogate models developed need to be updated accordingly and this is when ANNs are very useful as they can update themselves.

[0035] Results from the method may be output as a reporting value, for example, % of life consumed for each component. In another embodiment, a life estimation value for the gas turbine engine based on the life consumed for each component may be reported. A change in the operating parameters of the gas turbine engine may be implemented based upon these life consumption numbers. A variety of operating parameters that affect performance of the engine may include, for example, initiating a shutdown, reducing the load, and reducing the rotational speed of the engine. Such a

change may be monitored via the sensors and controls on the turbine engine, for example, monitoring the exhaust temperature of the turbine engine.

[0036] A system for estimating a remaining life of a gas turbine component in an operating gas turbine engine is also provided. Fig. 3 illustrates an embodiment of such a system. The illustrated embodiment shows a simplified depiction of a typical gas turbine power plant 1 with a generator 2 supplying a plant electric load 3. The generator 2 is driven by a shaft 4 powered by a gas turbine engine 5. The gas turbine engine 5 is itself comprised of a large number of separate components, including a compressor 6, a combustion section 7, a turbine 8, and perhaps, a set of adjustable inlet vanes 9. Fuel is supplied to the combustion section 7 via a valve 10. In order to maintain acceptable operation of the gas turbine power plant 1, a plurality of sensors 11 are used to monitor the operation of the various components, passing the measured sensor readings to a separate control module 12. The control module 12 may be co-located with the gas turbine power plant 1, or may be off-site from the turbine itself. As described above, the various sensors 11 measure many different parameters on the gas turbine engine 5, for example, measuring conditions such as temperature, pressure, rotation, vibration, etc. The control module 12 receives inputs from the sensors 11 and transmits control signals to valves, motors, and actuators known in the art. The control module 12 may include at least one processor.

[0037] The control module 12 may be in communication with a database 13 in order to store the sensor readings. The acquired data may add up to several terabytes and thus requires a secure, organized database where the data may be easily stored and queried in real time.

[0038] The sensor database 13 may be linked to a life estimation and control system comprising an analysis processing platform 14, comprising at least one processor 18 and some form of memory 16. The processing platform 14 outputs reporting values to a display 20 such as a visualization dashboard. Furthermore, the processing platform 14 may send control signals to the control module 12 based on the life estimation analysis to change a physical parameter of the gas turbine engine 5 such as initiating a shutdown, reducing a load, etc.

[0039] The above described method may be implemented by program modules that are executed by a computer. Generally, program modules include routines, objects, components, data structures and the like that perform particular tasks or implement particular abstract data types. The term ‘program’ as used herein may connote a single program module or multiple program modules acting in concert.

[0040] The disclosed method of estimating the current state of life consumption for a component of a gas turbine engine as well as the corresponding system accomplishes a fast and accurate method of life estimation for the components as well as the system as a whole. Furthermore, the proposed method may enable a system to have a higher reliability at a lower or comparable cost. Additionally, the proposed method of estimating life consumption of components as well as the systems that the components make up may apply to a variety of commercial applications, not only the described embodiment of the gas turbine engine.

[0041] While embodiments of the present disclosure have been disclosed in exemplary forms, it will be apparent to those skilled in the art that many modifications, additions, and deletions can be made therein without departing from the spirit and scope of the invention and its equivalents, as set forth in the following claims.

What is claimed is:

1. A method for estimating a life consumption for a component of a gas turbine engine, comprising:
 - providing a gas turbine component to be assessed on the gas turbine engine;
 - acquiring and storing 100 data relative to the current operating condition of the component during operation of the gas turbine engine;
 - validating 200 the acquired data; and
 - estimating 300 the life consumption of the component based on the acquired data.
2. The method as claimed in claim 1, the acquiring 100 comprising acquiring data, via sensors, indicative of the operating condition of the component during operation of the gas turbine engine.
3. The method as claimed in claim 2, wherein the acquired data is selected from the group consisting of vibration measurements, pressure measurements, temperature measurements, shaft speed measurements, corrosion measurements, ambient condition measurements, and combinations thereof.
4. The method as claimed in claim 1, the acquiring 100 comprising providing discrete historical data relative to the gas turbine component.
5. The method as claimed in claim 1, wherein the validating 200 includes filtering acquired data in order to remove errors and detecting sensor anomalies and biases in the filtered data.
6. The method as claimed in claim 1,
 - wherein the estimating 300 includes creating a surrogate model comprising a physical model of the gas turbine component, and
 - wherein the surrogate model is adjusted utilizing a plurality of correction factors derived from operational data.

7. The method as claimed in claim 6, wherein the physical model is created by:

determining cyclic damage utilizing equivalent starts, and
determining the equivalent life consumed utilizing equivalent baseload hours.

8. The method as claimed in claim 7, wherein determining cyclic damage utilizing equivalent starts is calculated using the equation:

$$N_{ES}^{cg} = \sum_{i=1}^{N_{cycles}} F_{transient}^{(ES,cg)}$$

Where:

N_{ES}^{cg} is total number of cycles of all types (including start-stop and loading sub-cycles),

cg refers to specific component or component group, and

$F_{transient}^{(ES,cg)}$ refers to the transient event start factor.

9. The method as claimed by claim 8, wherein determining the equivalent life consumed utilizing equivalent baseload hours is calculated using the equation:

$$LC_m^{total} = F_m^{online} \sum_{i=1}^{N_{online}} LC_{online} \Delta t_{online} + F_m^{offline} \sum_{i=1}^{N_{offline}} LC_{offline} \Delta t_{offline} + F_m^{start} \sum_{i=1}^{N_{start}} LC_{start} \Delta t$$

Where:

LC_m^{total} is the total life consumed for a damage mechanism 'm',

LC_{online} , $LC_{offline}$, LC_{start} are life consumed during online, offline and start conditions determined by laboratory tests and physics-based modeling as a % of useful life,

Δt are intervals of operation and

F_m^{online} , $F_m^{offline}$ and F_m^{start} are the plurality of correction factors

developed using machine learning techniques to operational data as applied to specific field conditions of individual machines.

10. The method as claimed in claim 9, wherein the damage mechanism is selected from the group consisting of corrosion, erosion, oxidation, hot corrosion, creep, high cycle fatigue, and off-line spinning.

11. The method as claimed in claim 9, wherein the plurality of correction factors are developed utilizing machine learning techniques in which the machine learning techniques automatically determine and learn functional relationships between component conditions and remaining life.

12. The method as claimed in claim 11, further comprising interrogating the surrogate model in real-time to determine cumulative life consumption of the component.

13. The method as claimed in claim 12, further comprising predicting life consumption of a gas turbine component utilizing the surrogate model.

14. A method for assessing the current state of an operating gas turbine, comprising:

performing the method as claimed in claim 1 for each of a plurality of gas turbine components cooperatively integrated on the operating gas turbine; and

assessing 400 the current state of the engine using a statistical combination of the life consumption of each of the plurality of gas turbine components.

15. The method as claimed in claim 14, the acquiring 100 comprising acquiring data, via sensors, indicative of the operating conditions of each component during operation of the gas turbine engine.

16. The method as claimed in claim 14, wherein the validating 200 includes filtering acquired data in order to remove noise and removing random errors from the filtered data.

17. The method as claimed in claim 14, further comprising predicting 500 life consumption of the gas turbine based on life consumption of each component and the operating conditions of gas turbine.

18. The method as claimed in claim 14, further comprising changing a physical operating parameter of the gas turbine in response to the assessment 400 of the current state of the gas turbine.

19. The method as claimed in claim 18, wherein the physical operating parameter of the gas turbine comprises at least one of the group consisting of initiating a shutdown, changing a load, and reducing the rotational speed of the gas turbine.

20. A system for estimating life consumption of a plurality of a gas turbine components in an operating gas turbine engine 5, comprising:

- a database 13 used to store acquired data and configured to be queried in real time to access the stored acquired data;

- a processing platform 14 including a program storage device and a first processor 18, the processing platform in communication with the database 13, the program storage device embodying in a fixed tangible medium a set of program instructions executable by the processor 18 to perform the method steps of:

- acquiring and storing data relative to the current operating condition of the components during operation of the gas turbine 5,

- validating the acquired data, and

- determining life consumption of each component based on the acquired data;

- a control module 12 including a second processor communicatively connected to the processing unit 14 and configured to change a physical parameter of the gas turbine 5 based on life consumption of the components; and

- a display 20 configured to display the determined life consumption of each component.

1/2

FIG. 1

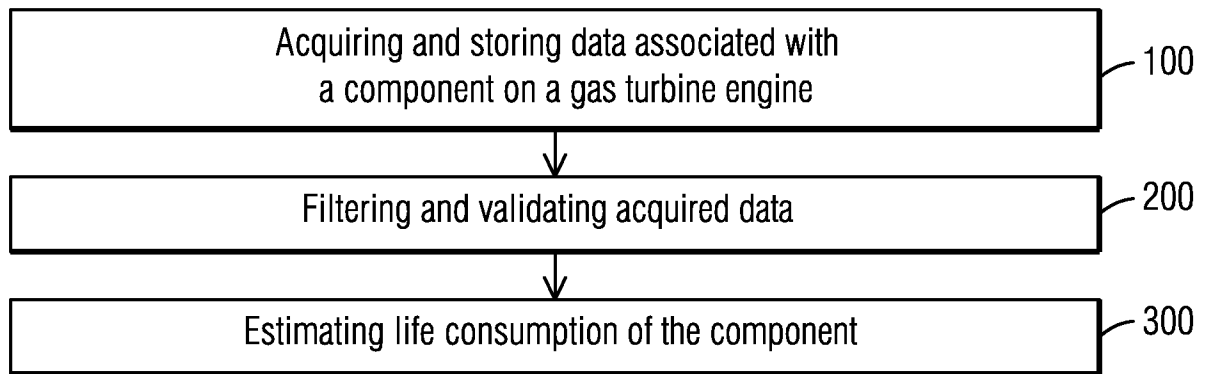


FIG. 2

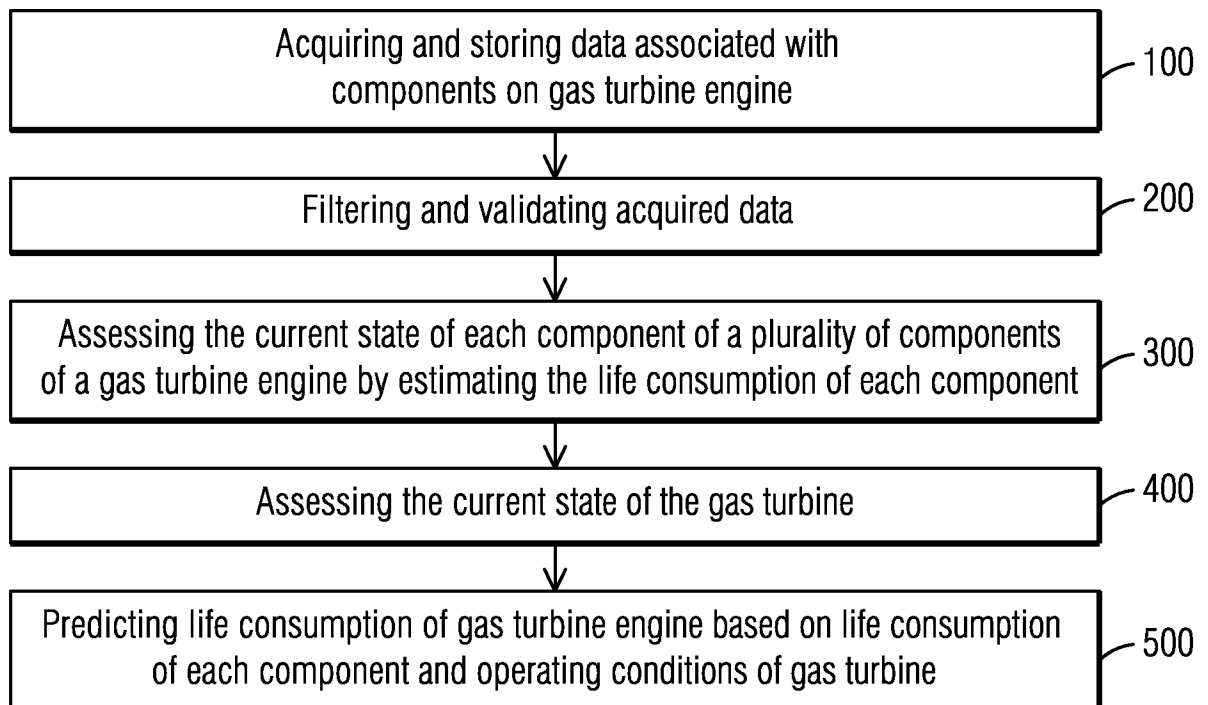
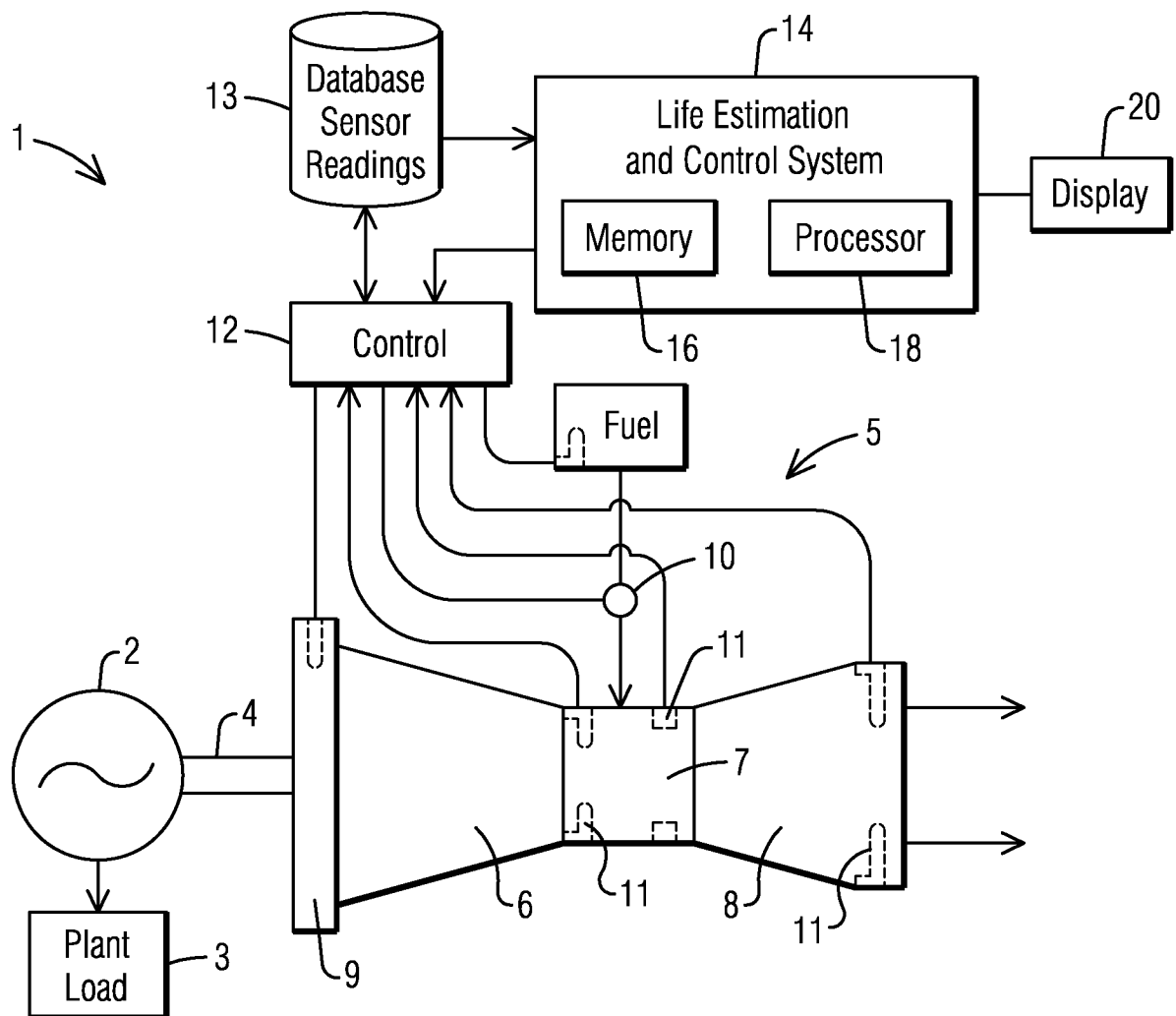


FIG. 3



INTERNATIONAL SEARCH REPORT

International application No
PCT/US2017/045685

A. CLASSIFICATION OF SUBJECT MATTER
INV. G05B23/02
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G05B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2013/191595 A1 (GKN AEROSPACE SWEDEN AB [SE]; NILSSON MIKAEL [SE]; NORDER DANIEL [SE]) 27 December 2013 (2013-12-27)	1-6, 14-17
Y	page 1 - page 14	7,18-20
Y	US 2008/208487 A1 (GOEBEL KAI FRANK [US] ET AL) 28 August 2008 (2008-08-28) page 1	18-20
Y	EP 2 821 873 A2 (BELL HELICOPTER TEXTRON INC [US]) 7 January 2015 (2015-01-07) paragraph [0001] - paragraph [0013]	7
A	WO 2013/191593 A1 (GKN AEROSPACE SWEDEN AB [SE]; ANDERSSON MAGNUS [SE]; WAENMAN FREDRIK []) 27 December 2013 (2013-12-27) the whole document	1,20



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

16 October 2017

Date of mailing of the international search report

20/10/2017

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2
NL - 2280 HV Rijswijk
Tel. (+31-70) 340-2040,
Fax: (+31-70) 340-3016

Authorized officer

Kuntz, Jean-Marc

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/US2017/045685

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2013191595 A1	27-12-2013	EP 2862034 A1 JP 2015529765 A US 2015234365 A1 WO 2013191595 A1	22-04-2015 08-10-2015 20-08-2015 27-12-2013
US 2008208487 A1	28-08-2008	NONE	
EP 2821873 A2	07-01-2015	EP 2821873 A2 US 2015007666 A1	07-01-2015 08-01-2015
WO 2013191593 A1	27-12-2013	EP 2862033 A1 JP 6069498 B2 JP 2015522747 A US 2015234951 A1 WO 2013191593 A1	22-04-2015 01-02-2017 06-08-2015 20-08-2015 27-12-2013